

My Project

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Chapter 1

Introduction

1.1 Purpose

This document is meant for users of Arxtron's Allen Bradley PLC Tester.

It is intended to demonstrate how it can be used and configured, where to find specific features, and what indicators/outputs mean.

Chapter 2

Operational Modes

This station has two main operational modes.

- 1. Manual Mode** All instruments and actuators are controlled through manual button presses on the [Manual Operation](#) page. Only [Administrator](#) users have access to this page.
- 2. Auto (Production) Mode** Can only be entered when the station is [homed](#) and all [faults](#) cleared. This mode allows the station to act automatically based on sensor and upstream/downstream inputs.

The [Main Interface](#) features some buttons with multi-state indicators. They are shown as grey buttons with an outer indicator boarder. The below table goes over the general meaning behind these indicators.

The light towers on the station also reflect the indicator for the **Auto** button.

Table 2.1 Indicator Meaning

Color	Meaning
Grey	Inactive
Blue	Ready/Interlocks met
Green	Active/OK
Yellow	In progress
Red	Fault/Not OK

Chapter 3

User Interface

3.1 Main Interface

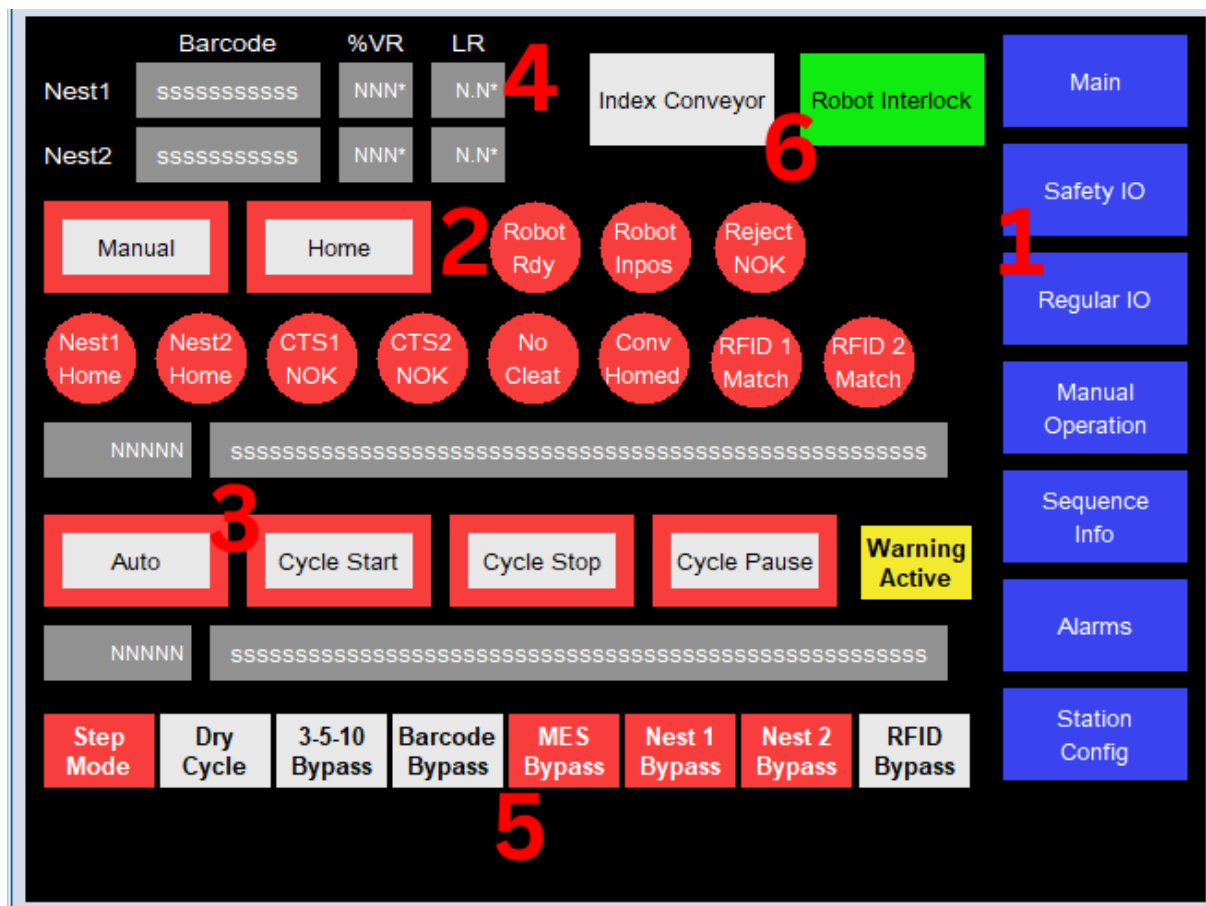


Figure 3.1 Main User Interface

1. [Navigation Buttons](#)
2. [Station Homing](#)
3. [Mode and Cycle Controls](#)
4. [Test Results](#)

5. [Modes and Bypasses](#)
6. [Index Conveyor and Robot Interlock](#)

The image shows all elements. The Robot Interlock button is only shown when an [Administrator](#) is logged in.

3.1.1 1. Navigation Buttons

These buttons are used to navigate between the different screens.

1. [Main Interface](#)
2. [Safety IO](#)
3. [Regular IO](#)
4. [Manual Operation](#)
5. [Sequence Info](#)
6. [Alarms](#)
7. [Station Config](#)

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3.1.2 2. Station Homing

The station needs to be in the homed state prior to entering Auto Mode. The homing sequence is started by pressing the **Home** button while in Manual Mode. The homing is considered complete when all indicators are green. The homing sequence has a hard time out of 2 minutes.

RFID Fixture Check Once both nest cylinders are in the home position, each nest fixture is at its RFID head. The homing sequence then scans the fixture RFID tag of Nest 1 followed by Nest 2 and compares each to the expected fixture value for the current module type (Short or Long) received from the previous station.

- If the tag matches, homing continues to the next step.
- If the tag does not match, homing stops and a [RFID mismatch fault](#) is raised for that nest. Install the correct fixture, or write the correct value to the tag through the [RFID Scanner](#) controls, then reset the fault and re-run homing.

The RFID fixture check only runs as part of the homing sequence. The fixture check is cleared, and the station must be homed again before entering Auto Mode, when:

- The E-Stop is pressed
- The module type received from the previous station changes

When changing over to a different module type, make sure the new module type has been received from the previous station before pressing **Home**.

Table 3.1 GUI Elements

Element	Purpose
Home (Temporary Button)	Starts the homing sequence. Shows Homing with a yellow border while the homing sequence is running, and Home with a green border once the station is homed
Robot Rdy (Indicator)	The ib_Ready signal is set to high from the robot
Robot Inpos (Indicator)	The robot is in the rest (barcode) position
Reject NOK (Indicator)	The reject bin is not present
Nest1 Home (Indicator)	The cylinder for nest 1 is in the home position
Nest2 Home (Indicator)	The cylinder for nest 2 is in the home position
CTS1 NOK (Indicator)	Leak tester 1 previous results not cleared
CTS2 NOK (Indicator)	Leak tester 2 previous results not cleared
No Cleat (Indicator)	Conveyor cleat is not in position
Conv Homed (Indicator)	Conveyor has at least N empty spots
RFID 1 Match (Indicator)	The nest 1 fixture RFID tag matches the expected value
RFID 2 Match (Indicator)	The nest 2 fixture RFID tag matches the expected value

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3.1.3 3. Mode and Cycle Controls

Table 3.2 GUI Elements

Element	Purpose
Manual (Temporary Button)	Changes the PLC to Manual Mode
Auto (Temporary Button)	Changes the PLC to Auto Mode
Cycle Start (Temporary Button)	Starts the production sequence. Can only be started when Auto Mode is active
Cycle Stop (Temporary Button)	- Stops the production sequence - Held (3s) force stops the production sequence - Cycle Stop is automatically triggered when a fault occurs during sequence run - Cycle Stop is unset when Cycle Start is pressed - The production sequence will attempt to start at step 1000 when resumed
Cycle Pause (Temporary Button)	- Pauses the production sequence - Cycle Pause is unset when Cycle Start is pressed - The production sequence will attempt to start at the paused step when resumed
Warning Active (Banner)	Shown when a warning is active

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3.1.4 4. Test Results

Displays the last scanned barcode for each nest as well as the % Volume (VR) and Leak Rate (LR) test results.

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3.1.5 5. Modes and Bypasses

All buttons found here other than Next Step are toggle buttons. All bypasses should be toggled on prior to pressing Cycle Start.

Table 3.3 Modes and Bypasses

Button	Purpose
Step Mode/Next Step	Puts the production sequence in Cycle Pause and allows for manual sequence stepping
Dry Cycle	Production sequence will operate as if parts are present
3510 Bypass	Bypasses faults caused by 3 5 10 rules
Barcode Bypass	Bypasses barcode scanning in production sequence
MES Bypass	Bypasses all MES related checks
Nest1/2 Bypass	Bypasses selected nest during production sequence
RFID Bypass	Bypasses the RFID fixture check during homing

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3.1.6 6. Index Conveyor and Robot Interlock

Index Conveyor: Indexes the conveyor 1 full position. Only displayed in Auto Mode, for all users. See [Index Conveyor \(Auto\)](#).

Robot Interlock: Allows the robot to be run manually. Only displayed when an [Administrator](#) is logged in, in both Manual and Auto Mode.

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3.2 Main Interface (Auto Mode)

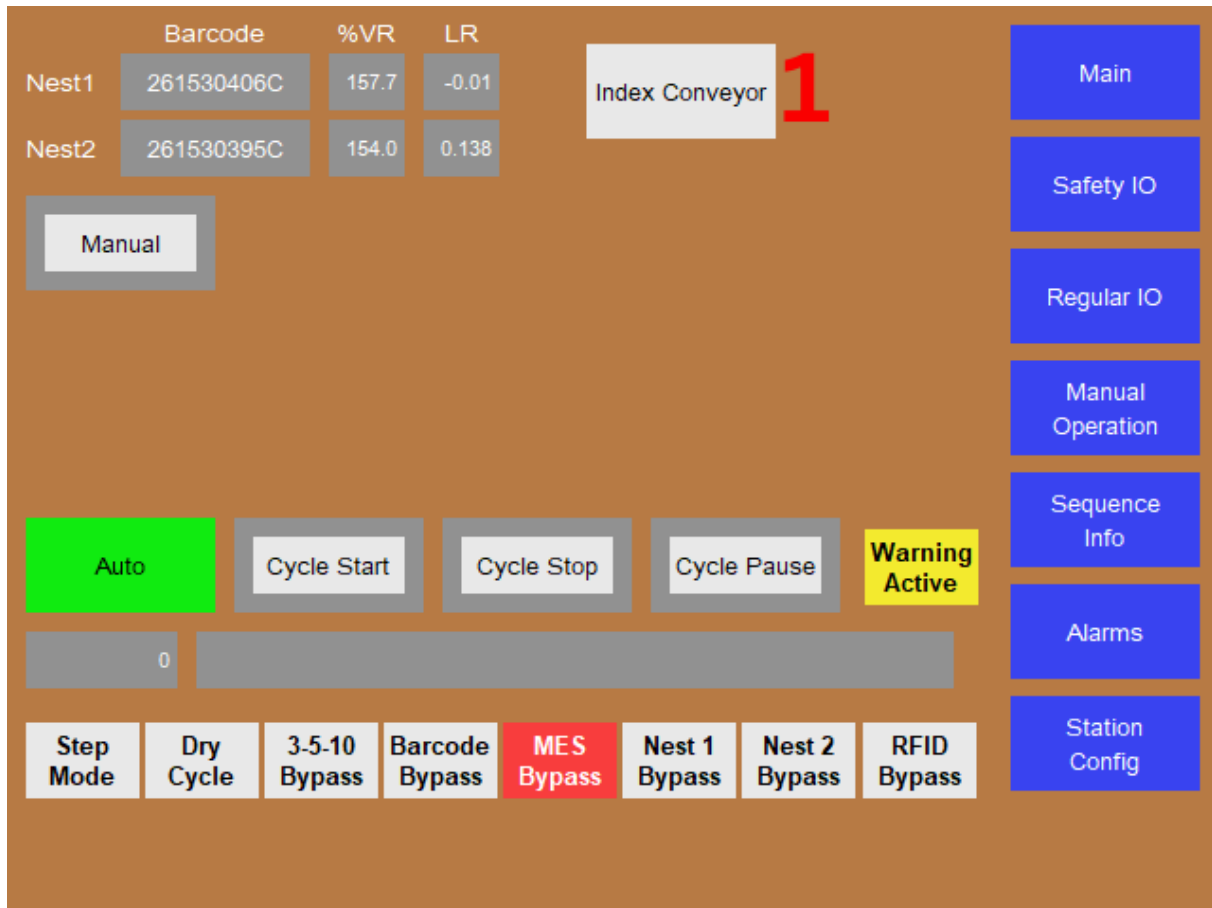


Figure 3.2 Main User Interface (Auto Mode)

1. Index Conveyor (Auto)

The Auto button is shown green while the station is running in Auto Mode. Homing controls are hidden in Auto Mode.

3.2.1 1. Index Conveyor (Auto)

Allows operators to index the conveyor manually while in Auto Mode. Available to all users.

3.3 IO Pages

All IO pages follow a similar layout of LEDs and labels

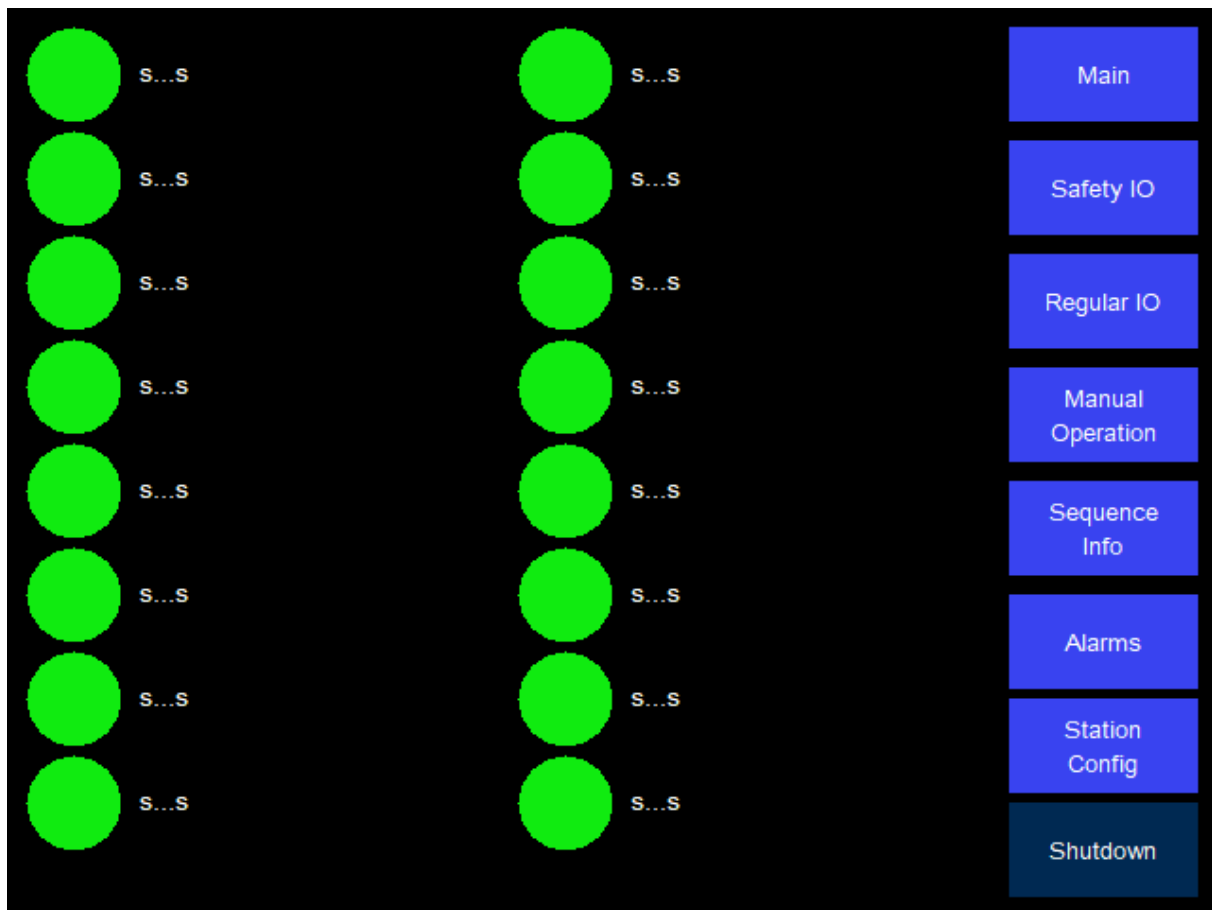


Figure 3.3 IO Indicators

3.3.1 Safety IO

The Safety IO page only displays the status of all safety inputs and outputs.

3.3.2 Regular Inputs

The Regular Input pages display the status of all input signals as well as its global trigger count. These counts are reset when pressing the LED. These counts are only used for debug purposes and are not and should not be used in logic.

3.3.3 Regular Outputs

The Regular Output page displays the status of all output signals. The LEDs act as toggle buttons, but do not override PLC logic. As such, most signals cannot be toggled from this screen.

3.4 Sequence Info

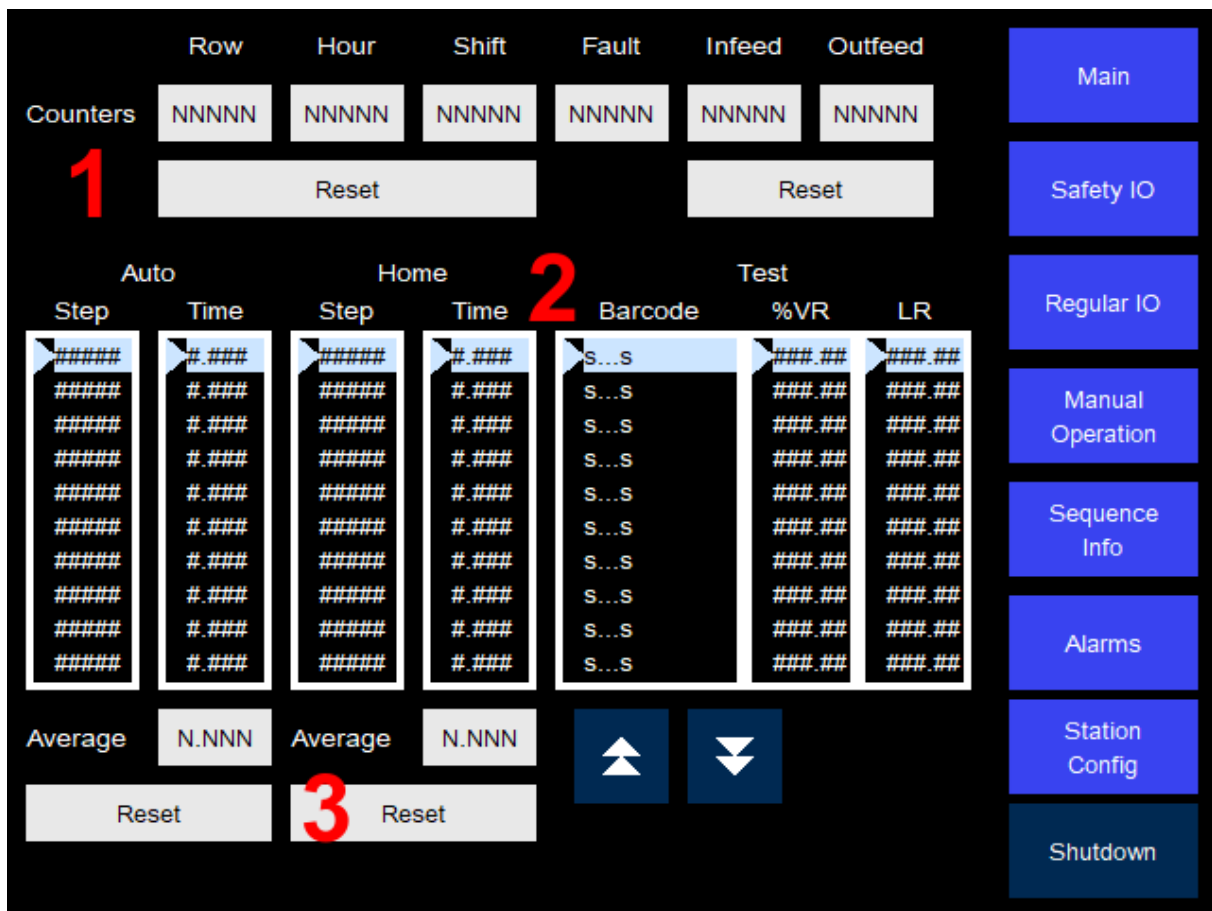


Figure 3.4 Sequence Info

Reset buttons are only visible when an [Administrator](#) is logged in.

1. [Counters](#)
2. [History](#)
3. [History Reset](#)

3.4.1 1. Counters

Table 3.4 Counters

Counter	Purpose	Reset
Row	Number of failed units in a row	Non-consecutive fail
Hour	Number of failed units per hour	New hour
Shift	Number of failed units per shift	Shift change
Fault	Number of faults triggered, concurrent faults may only count once	Shift change
Infeed	Number of units picked up from load nest	Button
Outfeed	Number of units unloaded to conveyor	Button

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3.4.2 2. History

Sequence lists display the step history. Test lists display the test history. Navigation buttons found at the bottom will navigate only the active list. Related lists are not synced in terms of positions and need to be individually aligned via navigation buttons.

All lists are in reverse order, where the latest data populate the top most entry. If an array is not fully populated, the top entries of the list may be empty.

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3.4.3 3. History Reset

Average indicates the average cycle time. The reset buttons will reset all sequence related data.

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3.5 Alarms

View and reset faults and warnings.

3.6 Manual Operation

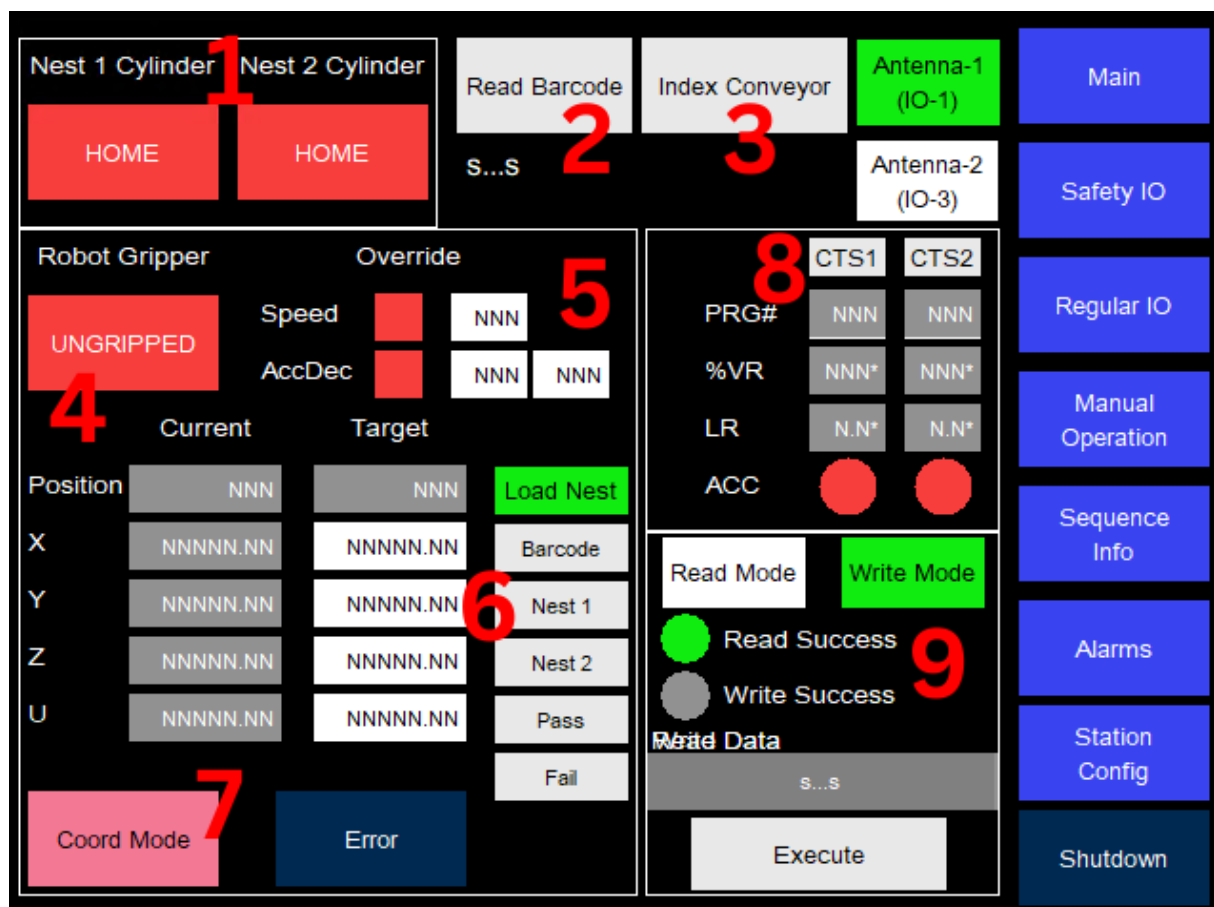


Figure 3.5 Manual Operation

All controls shown here are only available in Manual Mode.

1. [Nest Cylinders](#)
2. [Barcode Scanner](#)
3. [Conveyor Indexing](#)
4. [Robot Gripper](#)
5. [Robot Override](#)
6. [Robot Movement](#)
7. [Robot Position/Coord Mode](#)
8. [Leak Testers](#)
9. [RFID Scanner](#)

3.6.1 1. Nest Cylinders

Manual controls for nest cylinder position.

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3.6.2 2. Barcode Scanner

Manual trigger for barcode scanner.

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3.6.3 3. Conveyor Indexing

Manual indexing of conveyor. Button press will index conveyor 1 full position. Jogging not possible.

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3.6.4 4. Robot Gripper

Manual toggle of robot gripper. Gripping without a DUT will cause a gripper fault.

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3.6.5 5. Robot Override

Override default robot parameters based on entered values. Overrides persist in Auto Mode.

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3.6.6 6. Robot Movement

Toggle buttons or Target Position sets robot movement target. Press Start to execute movement. Moving to Load Nest requires upstream interlock.

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3.6.7 7. Robot Position/Coord Mode

Toggles robot movement between Position mode and Coord mode. The same button switches between the two modes and is only displayed in Manual Mode.

- **Position mode:** robot moves to the selected [target position](#).
- **Coord mode:** robot moves using coordinate input. Robot movement unpredictable and not fully safe. Use with caution.

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3.6.8 8. Leak Testers

Manually set leak tester program number and trigger tests.

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3.6.9 9. RFID Scanner

Manual read and write of the fixture RFID tags. Read Mode is available to all users. Write Mode is only shown when an [Administrator](#) user is logged in.

1. [Antenna Selection](#)
2. [Read Mode / Write Mode](#)
3. [Status Indicators](#)
4. [Tag Data](#)
5. [Execute](#)

3.6.9.1 9.1 Antenna Selection

Selects which RFID head the command is sent to.

- **Antenna-1 (IO-1):** Nest 1 RFID head
- **Antenna-2 (IO-3):** Nest 2 RFID head

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3.6.9.2 9.2 Read Mode / Write Mode

Selects the command to run on Execute.

- **Read Mode:** Reads the data stored in the tag
- **Write Mode:** Writes the entered data to the tag. Only visible when an Administrator is logged in

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3.6.9.3 9.3 Status Indicators

- **Read Success:** Turns green for 3 seconds after a successful read
- **Write Success:** Turns green for 3 seconds after a successful write

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3.6.9.4 9.4 Tag Data

In Read mode, displays the 24 character value read from the tag after Execute is pressed. In Write mode, pressing the box opens a keyboard to enter the 24 character value to write.

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3.6.9.5 9.5 Execute

Runs the selected Read Mode / Write Mode command on the selected antenna.

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3.7 Station Config

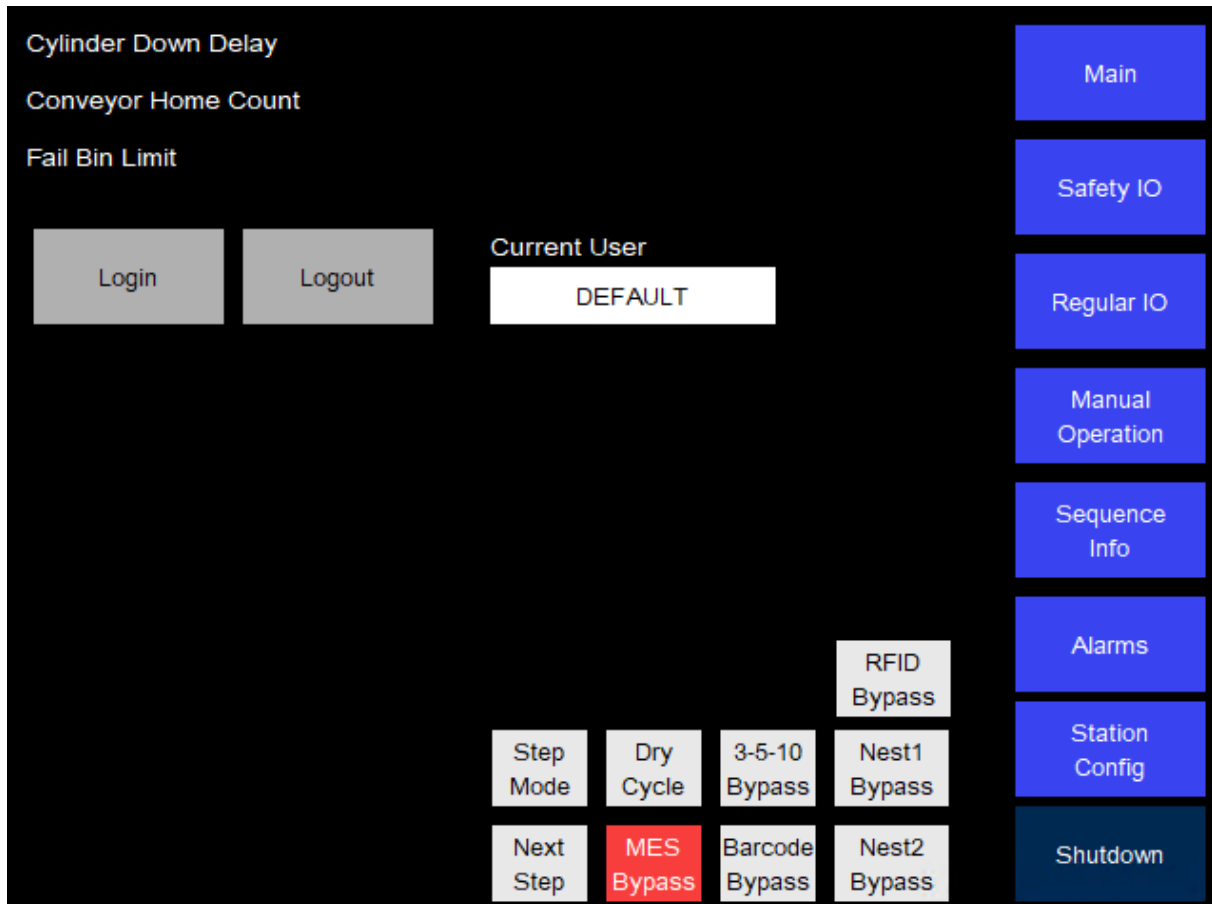


Figure 3.6 Station Config

Cylinder Down Delay: 500

Conveyor Home Count: 0

Fail Bin Limit: 10

Login Logout

Current User: Admin

Auto Mode CTS Test: 2 Tall

CapShuffle CapRobot CapFail

CapActuators CapCTS CapCylinders

CapPass

Step Mode Dry Cycle 3-5-10 Bypass Nest1 Bypass

Next Step MES Bypass Barcode Bypass Nest2 Bypass

RFID Bypass

Main Safety IO Regular IO Manual Operation Sequence Info Alarms Station Config Shutdown

Figure 3.7 Station Config (Administrator)

1. [User Authentication](#)
2. [Station Parameters](#)
3. [Auto Mode CTS Test](#)

System-wide [modes and bypasses](#) are also available at the bottom of this page.

3.7.1 1. User Authentication

Used for logging into higher level access users. **Current User** shows the logged in user. Users in the **Administrator** group will see additional controls on various pages. Authenticated users will be logged out after 10 minutes of idle time.

Table 3.5 Users

User	Password	Group
MAGNA	MAGNA	Administrators
Admin	Admin	Administrators
DEFAULT		

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3.7.2 2. Station Parameters

Only editable when an Administrator is logged in.

Cylinder Down Delay: Time in ms delay between triggering the cylinder down sensor and starting the leak tester tests. **Conveyor Home Count:** Number of empty pockets required when homing conveyor. **Fail Bin Limit:** Number of DUTs registered to the fail bin before it is prompted for emptying.

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3.7.3 3. Auto Mode CTS Test

Only visible when an Administrator is logged in.

CTS: Program number loaded on the leak testers. **DUT Type:** Type of DUT being tested, **Tall** or **Short**.

The Cap buttons (CapShuffle, CapRobot, CapFail, CapActuators, CapCTS, CapCylinders, CapPass) select capability test modes.

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Chapter 4

Faults and Warnings

Table 4.1 Faults

Category	Cause	Recovery
Utility	Lack of power/air/safety	Address according to message
EIP connection	Instrument off/disconnected or communication error	Check instrument EthernetIP settings
EStop fault	Safety program block fault	Manually trigger reset in program
Instrument operation	Various causes	Check instrument is functioning correctly and all interlocks are met
RFID mismatch	• Cylinder 1 RFID not matching expected value. • Cylinder 2 RFID not matching expected value.	Install the correct fixture in the affected nest, or write the correct value to the fixture tag through the RFID Scanner controls. Reset the fault and re-run homing
Sequence	Unexpected behavior/state	Check step history and resolve based on expected behavior

Table 4.2 Warnings

Category	Cause	Recovery
Mode active	Special mode or override active	Turn off mode and reset faults
Sequence timeout	Sequence does not progress to next step within certain time	Check step history and resolve accordingly
Instrument operation	Various causes	Check instrument is functioning correctly and all interlocks are met
Fail bin full	Fail bin counter reached	Remove and empty fail bin

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